

Gaurav Bhatia

Berlin, Germany | gauravbhatia.gb6@gmail.com | +49 152 11632128
Portfolio | GitHub | LinkedIn

Analytics Engineer / Junior Data Engineer

MSc Data Science candidate at GISMA, finishing December 2026, who builds the data layer that analysis depends on. Writes **SQL** (CTEs, window functions) in DuckDB and Snowflake, models **star schemas**, and automates validation: **45 checks** across 7 tables that traced two silent data losses. Schedules and ships pipelines with Python, Docker and GitHub Actions. Available full-time in Berlin from **December 2026**.

Skills

SQL & modelling	SQL (CTEs, window functions, views), DuckDB, Snowflake SQL, MySQL, SQLite, DWH / star-schema modelling, ETL processes, Python (pandas, NumPy)
Pipelines & delivery	ETL processes, data-quality validation (a 45-check suite), scheduled pipelines (APScheduler), REST APIs, FastAPI, Docker, GitHub Actions CI/CD, Git/GitHub, documented and reproducible pipelines
Cloud & platforms	Azure data platform (FutureLearn coursework: data engineering with Microsoft Azure), AWS, Snowflake, Docker, Kubernetes, Agile/Jira
Charts & reporting	Power BI / DAX, Plotly, Streamlit dashboards, automated Markdown/HTML KPI reports, Excel, Tally
Languages	English (C1), German (A2, improving), Hindi (native)

Experience

Data Science Intern Aug 2026 – Present
KnowGuard AI *Berlin, Germany*
- Build **RAG pipelines** and **multi-agent systems**, and define evaluation metrics and hold-out sets so model comparisons are reproducible rather than anecdotal
- Ran a **paired A/B test on AWS** comparing a fine-tuned model against the review arm: **+0.0318 AUPRC** (95% CI 0.022–0.0418), below the pre-set 0.05 practical-effect threshold

Operations & Supply Chain Analyst Mar 2024 – Jul 2025
Troika International Pvt. Ltd. *Panchkula, India*
- Built **Power BI dashboards across 3 divisions**, turning operational data into clear views for non-technical stakeholders
- Worked directly with operations and supply-chain stakeholders to turn recurring questions into **standing reports**
- Monitored inventory and demand across **700+ vendors** and **500+ SKUs** in Tally and Excel

Business Manager Nov 2015 – Jan 2024
Bhatia Traders *India*
- Ran a product business end to end for **8 years**, owning purchasing, pricing, stock and supplier terms across **20+ suppliers**

Education

GISMA University of Applied Sciences Potsdam, Germany
MSc Data Science, AI & Digital Business *Sep 2025 – Dec 2026 (expected)*
- Thesis: the US–Iran conflict's impact on German day-ahead electricity prices, using event study, Granger causality and Chronos/XGBoost forecasting
- Relevant modules: Methods of Prediction (M507), Big Data Analytics (M508)

University College of Engineering, Patiala *B.Tech, Civil Engineering, 2011 – 2015*

Selected Projects

BI Data-Quality Pipeline *Python, pandas, NumPy* Source
- Automated **45 validation checks** (missing values, duplicates, referential integrity, outliers, row counts) across a 7-table star schema; **41/45 passed (91.1%)**
- Traced two silent data losses (614 payment rows, 551 review rows) to root cause and documented fixes in an auto-generated report

Toll Traffic Reporting and Anomaly Suite *DuckDB, SQL, Isolation Forest, APScheduler, Jinja2* Source
- Built **SQL reporting** (CTEs, window functions, RANK), **Z-score and Isolation Forest** anomaly detection, and a scheduled weekly HTML KPI report over German highway toll data
- Covers 6 months of hourly data across 5 autobahn segments with 4 standing reports; the Monday 08:00 report is **fully automated**

Solar Demand Analytics *DuckDB SQL, Open-Meteo, SMARD, Power BI* Source
- Modelled German solar generation and weather in **DuckDB SQL** (window functions, views) and exported to **Power BI**: summer output 4x winter, cloud cover above 75% cuts generation ~40%

Payments BI Dashboard *Power BI, DAX, Python, star schema* Source
- Built a 3-page **Power BI** dashboard over **100K payment transactions** ($15.99M revenue) covering BNPL share, delivery performance and revenue mix with **DAX** measures

Job-Search Automation: agent harness *Scheduled agents, shared database, Python, GitHub* Private repo

- Built a versioned system of **four scheduled agent tasks** (sourcing, outreach, daily report, LinkedIn drafts) that hand work to each other through a **shared database**
- Sourced roles from ATS and Arbeitsagentur feeds, government portals and job boards across **12 countries**, plus funding news for startups
- Governed it as an **agent harness**: **37 standing rules** with an ownership map, a **validator that gates every run** and issues receipts, and a regression test built on a real report

RAG Retail Agent *LangChain, FAISS, FastAPI, Docker, Kubernetes* Source

- Built a **LangChain + FAISS** retrieval pipeline over product data behind a **FastAPI** service, with **GitHub Actions to Docker Hub to Kubernetes** CI/CD

Odd-Jobs Aggregator for Students *Multi-source ingestion, dedup, LLM drafting* Private repo

- Built a **multi-source aggregator** collecting listings posted in the last **24 hours** across Kleinanzeigen, Craigslist and further pluggable sources, for students who cannot search them by hand
- Generates the message and a tailored CV per listing, then applies once the user approves a shortlist. **Human-in-the-loop** by design

MSc Thesis: Conflict Shocks in the German Power Market *Python, Chronos, XGBoost, Granger causality* Source

- Built a reproducible pipeline answering **5 research questions** on how the 2026 US–Iran conflict moved German day-ahead electricity prices, benchmarked against the 2022 energy shock
- Combined **event study**, **Granger causality** and ablation tests with **Chronos** and XGBoost forecast arms to identify the transmission channel

Inventory Alert System *MySQL (CASE, JOINs, UPDATE)* Source

- Built MySQL queries that raise automated **low-stock alerts at a 20% threshold** with reorder recommendations for a sauces and condiments business

Certifications

- **Introduction to Data Engineering with Microsoft Azure 1 & 2**, FutureLearn, Sep 2026
- **Introduction to Cloud Infrastructure: Describe Cloud Concepts**, Microsoft, Jul 2026
- **Data Analytics Job Simulation**, Deloitte, issued by Forage, Jul 2026